



Does Performing Resistance Exercise with a Partial Range of Motion at Long Muscle Lengths Maximize Muscle Hypertrophic Adaptations to Training?

Enrique N. Moreno¹ · Wayne A. Ayers-Creech¹ · Selena L. Gonzalez¹ · Holly T. Baxter² · Samuel L. Buckner¹ 

Received: 4 April 2024 / Accepted: 12 June 2024
© Beijing Sport University 2024

Abstract

Background Over the past few years, the effects of manipulating range of motion (ROM) on muscle hypertrophy has garnered a considerable amount of attention within the scientific community. When seeking to maximize muscle hypertrophy, it has previously been suggested that individuals should perform a given exercise over the largest possible degrees of movement (i.e., full ROM), however, recent review papers have suggested that performing a partial ROM at long muscle lengths (LML) could potentially promote superior hypertrophy compared to other ROM configurations.

Purpose We sought to examine the evidence for such suggestions as well as possible physiological mechanisms underpinning such phenomena.

Findings and Conclusion When assessing the literature, it appears that (1) there are not compelling data to support the suggestion that a partial ROM at LML is superior to full ROM, (2) it may be the case that a partial ROM at LML promotes greater distal hypertrophy when compared to a partial ROM at short muscle lengths (SML) but may promote comparable hypertrophy at more proximal sites, and (3) this phenomena (i.e., partial ROM at LML being the optimal ROM for hypertrophy) may not be generalizable to all muscle groups. Future research should seek to directly compare the different ROM configurations across a variety of exercises for all major muscle groups to understand whether an “optimal ROM for muscle hypertrophy” is dependent on both muscle group and exercise selection.

Keywords Range of motion · Muscle hypertrophy · Resistance training · Training variables

Introduction

Resistance training is an effective strategy for inducing skeletal muscle adaptations such as muscle hypertrophy (i.e., increases in the cross-sectional area (CSA) of muscle/muscle fibers [20]) in humans [34]. Extensive efforts have been made throughout the years to elucidate the effects of manipulating various training variables such as training volume (e.g., sets performed per muscle group) [15, 61], external load [37, 38], rest intervals [10, 60], etc., on muscle hypertrophy. In addition, the influence that joint range of

motion (ROM) has on modulating muscle hypertrophy has sparked a considerable amount of interest amongst the scientific community in recent years [22, 51, 52, 57, 70].

Range of motion has been dichotomized throughout the literature as performing exercise with either a full ROM (i.e., executing the largest degree of movement without any restrictions) or partial ROM (i.e., involving some limitations or restrictions in the degree of movement executed) [50]. However, partial ROM may be further categorized or discussed as it relates to what specific portion of the ROM is being performed. For example, using the concentric muscle action as reference, performing only the initial portion of the ROM has been termed as partial ROM at long muscle lengths (LML) where the muscle is thought to be more lengthened or stretched while performing only the final portions of the ROM has been termed as partial ROM at short muscle lengths (SML) where the muscle is suggested to be more shortened or contracted [21, 71].

✉ Samuel L. Buckner
slbuckner@usf.edu

¹ USF Muscle Lab, Exercise Science Program, University of South Florida, Tampa, FL, USA

² Holly Baxter LLC, Fort Lauderdale, FL, USA

Traditionally, it has been suggested that performing full ROM exercise may lead to a more favorable muscle hypertrophy response than otherwise may be achieved with partial ROM [50, 59]. However, recent review papers have proposed that full ROM training may only be superior to partial ROM when it is performed at SML, while hypertrophic responses may be similar, if not enhanced when performing partial ROM at LML [21, 71]. For example, Wolf et al. performed analyses to distinguish the effects of different partial ROMs (either partial ROM at SML or LML) in comparison to full ROM training for muscle hypertrophy and observed a modest benefit in favor of partial ROM at LML compared to full ROM (standardized mean difference = -0.28 ; 95% CI: $-0.81, 0.16$), while partial ROM at SML was slightly inferior to full ROM (standardized mean difference = 0.08 ; 95% CI: $-0.24, 0.42$), [71]. These findings led authors to conclude (1) that performing full ROM exercise may only be beneficial for muscle hypertrophy if it includes LML and (2) if the primary goal is to maximize muscle hypertrophy, avid gym goers should prioritize training with partial ROM at LML as opposed to other ROMs [71].

It is without standing that this area of research appears much more nuanced than is otherwise being portrayed to support such conclusions. For starters, there are only three studies to date that directly compare full ROM exercise to partial ROM at LML, one on the gastrocnemius [22] and two on the quadriceps [51, 70], with somewhat inconsistent findings. Moreover, much of the literature in general that has investigated manipulating ROM for the same given exercise(s) on muscle hypertrophy outcomes has been conducted on the quadriceps [5, 26, 35, 36, 51, 67, 70], with a few studies on the biceps [52, 54, 57], only one study on the triceps [14], and only one study on the gastrocnemius [22]. Because of such limited data, where many studies are confined to the quadriceps muscles, it is unclear whether partial ROM at LML is indeed the optimal ROM to maximize hypertrophy and if so, whether these findings apply to all major muscle groups. Furthermore, the physiological mechanisms as to why performing a partial ROM at LML would lead to superior muscle hypertrophy than other ROMs is somewhat ambiguous in nature, with some proposed theories contradicting other lines of research within the resistance training literature (e.g., metabolite accumulation driving hypertrophic changes, advantage of eliciting greater muscle damage through performing a given ROM).

Therefore, the purpose of the present manuscript was to explore the evidence for the suggestion that performing exercise with partial ROM at LML is a potential method for maximizing muscle hypertrophy and the physiological mechanisms that may (or may not) underpin this phenomenon.

How Does ROM Affect the Hypertrophic Stimulus?

The underlying mechanisms as to how ROM may influence the hypertrophic stimulus to resistance exercise are not fully elucidated, however, several working theories have been presented including (1) the location and where a muscle is most active on the length-tension curve may dictate the magnitude of mechanical tension experienced when manipulating ROM or muscle length [48], and (2) the role that metabolic stress and hypoxia may play in augmenting muscle anabolism when manipulating ROM or muscle length [14]. Prior to a discussion on such theories, it may benefit the reader to provide a brief overview of the proposed molecular mechanisms regulating resistance training-induced muscle hypertrophy. For a more extensive breakdown of such mechanisms the reader is pointed to several eloquent review papers on the topic [30, 33, 68]. It is generally understood that mechanical tension imposed by muscular contraction is the primary mechanism regulating resistance training-induced muscle hypertrophy [30, 68]. It is suggested that force generated by muscle fibers are transmitted to tendons and bones either longitudinally spanning end-to-end of a muscle fiber or laterally from the sarcolemma to the extracellular matrix [64] through mechanosensory proteins. Subsequently, it is proposed that these mechanical signals are converted into biochemical signals that may regulate the control of skeletal muscle size [6, 56]. The biochemical signals are capable of increasing muscle size through the activation of mechanistic target of rapamycin complex 1 (mTORC1)-dependent and -independent pathways [18, 72] that potentiate increases in muscle protein synthesis (MPS). This is an important consideration given that muscle hypertrophy may only occur in the presence of net positive protein balance, which occurs when the rates of MPS exceed those of muscle protein breakdown (MPB) for a prolonged period of time [53, 55]. While it is posited that manipulating various training variables may alter the hypertrophic stimulus (via eliciting varying degrees of MPS following exercise), it appears that a chief component of hypertrophy-oriented protocols involves performing repeated contractions that require a high degree of voluntary effort that consequently necessitates high degrees of motor unit recruitment and muscle activation [19, 39].

Increased Mechanical Tension When Manipulating ROM?

Being that mechanical tension is suggested to be the primary mechanism regulating muscle hypertrophy following resistance training, it has been proposed that altering a muscle's length or ROM during exercise may change the degree

of mechanical tension experienced via active and passive elements [21, 48]. Using the length-tension curve, which depicts the relationship between tension and sarcomere length (i.e., the functional units of myofibrils) [13, 29], there are three regions relating to sarcomere length: the ascending limb relating to shorter muscle lengths, plateau portion relating to the resting muscle length, and descending limb relating to longer muscle lengths. Pertaining to the active force producing capability of skeletal muscle fibers, the ascending limb is often associated with increasing force production, the descending limb is associated with decreasing force production, and the plateau portion being the optimal length for force production [29]. However, when accounting for passive force production via components such as titin and muscle tendons, it is theorized that muscle fibers at long lengths may be associated with the highest degrees of overall tension produced, even greater than the optimal muscle length, due to the combination of active and passive force production [16, 17]. Such rationale has led many authors to suggest that training at LML should be the default approach for maximizing muscle hypertrophy [21, 71], as it would involve producing the highest degrees of mechanical tension. However, as pointed out in a recent review paper by Ottinger and colleagues [48], this phenomenon likely does not apply to all muscles. For example, muscles such as the triceps brachii and biceps brachii are less likely to experience greater degrees of mechanical tension from passive elements when training at LML as they are more active on the plateau portion and ascending limb, respectively, of the length-tension curve [48]. However, the quadriceps of which are more active on the descending limb may benefit from training at LML [48]. Therefore, if we use this concept, training exclusively at LML may only be sensible for certain muscles and it seems reasonable that other muscles may respond more favorably to other ROMs. However, it may be too early to conclude whether findings from studies investigating the effects of manipulating ROM can solely be explained via the length-tension curve theory.

Role of Metabolic Stress and Hypoxia When Manipulating ROM?

In addition to the possible relationship between ROM or muscle length and mechanical tension, some authors have suggested that the heightened metabolic stress and deoxygenation that occurs when manipulating ROM may contribute to greater muscle anabolism [14, 51]. Several studies have suggested that muscle oxygen command and substrate utilization is greater when performing isometric contractions at LML compared to SML [24, 44], inferring greater metabolite production (e.g., lactate, hydrogen ions, inorganic phosphate) as a result of ATP turnover. Furthermore,

Goto et al. [14] conferred that greater levels of hypoxia resulting from partial ROM training explained 49% of the variance for hypertrophic adaptations when performing triceps brachii exercise (study will be discussed below more in depth), while the lower levels of hypoxia induced by full ROM training explained little to no variance of changes in muscle hypertrophy (i.e., 3% variance). However, it must be noted that correlations may not be used to imply causation and other factors may have influenced these findings.

The role of metabolite accumulation in promoting hypertrophic adaptations has been discussed throughout several lines of work [49, 58, 62]. While metabolite production is augmented with muscle contraction [31, 66], we are unaware of studies that have been conducted in humans that have demonstrated any direct evidence that metabolites per se contain anabolic properties. Instead there is a growing consensus that metabolic stress is not a primary driver of muscle hypertrophy [30], but instead may indirectly play a role through augmenting muscle activation and motor unit recruitment [7]. Meaning, it seems as though enhanced metabolite accumulation and increased hypoxia may only promote greater muscle hypertrophy in the presence of sub-maximal protocols (e.g., performing exercise far from task failure) as illustrated by earlier repetition-matched studies in the blood flow restriction literature [28]. However, if exercise is performed to or near fatigue (necessitating high degrees of motor unit recruitment and activation) there appears to be no added benefits of metabolites or hypoxia [9, 27, 65]. Therefore, it is unlikely that metabolites or hypoxia play any significant role in muscle hypertrophy when manipulating ROM so long as studies prescribe exercise to or near fatigue.

The Effects of ROM on Muscle Hypertrophy

Studies Directly Comparing Partial ROM at LML to Full ROM

Despite several review papers stating partial ROM at LML may be superior to full ROM exercise [3, 21, 71], there is surprisingly very little data comparing the two ROMs head-to-head [22, 51, 70]. In recreationally trained individuals (i.e., having performed at least 1 lower body training session per week for the past 6 months prior to enrollment), Werkhausen et al. [70] demonstrated no difference in the hypertrophic response of the vastus lateralis when performing either full ROM (defined as a starting knee angle 90° and end knee angle 0° [0° being full knee extension]) or partial ROM at LML (defined as a starting knee angle 90° and end knee angle 81°) for the unilateral leg press exercise. In fact, however, when assessing the data, there were no meaningful

changes in muscle thickness, measured via B-mode ultrasound, for the vastus lateralis (60% the distance between the greater trochanter and the femoral lateral epicondyle) for either training condition, despite performing exercise thrice weekly for 10 weeks [70]. This may be explained by a variety of factors such as the training protocol which consisted of explosive, concentric-only training with the inclusion of inter-repetition rest intervals (i.e., 3–6 sets of 4–8 repetition maximum [RM]; programming was manipulated depending on the day and week [70]), that may have led to an inadequate amount of contraction-induced time under tension and overall mechanical tension (i.e., from passive and active elements) necessary to stimulate robust muscle hypertrophy. Furthermore, the authors only measured changes in muscle growth at one site of the vastus lateralis, and since it has been suggested that muscle growth occurs inhomogeneously across the muscle belly [69, 73], the methods of this study may have prevented the observation of meaningful changes in other regions.

Contrary to these findings, Pedrosa and colleagues [51] reported that performing bilateral leg extension with a partial ROM at LML (defined as a starting knee angle 100° and end knee angle 65°) resulted in superior increases in rectus femoris muscle CSA, measured via B-mode ultrasound, than a group performing the same exercise with a full ROM (defined as a starting knee angle 100° [0° being full knee extension] and end knee angle 30°). However, changes in vastus lateralis muscle CSA were not statistically different between groups. At the surface, these findings would seemingly support the length-tension curve theory because the quadriceps should be more responsive to training at LML [48, 63]. However, caution may be warranted with these results. One major limitation with the study design was the prescription of a standardized repetition and loading schemes that did not necessarily account for local differences in muscle endurance. For example, it was noted that individuals performed 3–6 sets (depending on the week) of 7 repetitions at 60% of 1RM, respective to each ROM. There is some concern when prescribing such standardized protocols (i.e., fixed number of repetitions at a distinct % of 1RM) as there is a large degree of variability as to the number of repetitions able to perform at a given percentage of 1RM and thus proximity to failure may be different across individuals [8]. Reason for concern is that some studies have suggested that performing exercise exclusively at LML results in significantly less time necessary to reach exhaustion compared to SML [11, 44], therefore, it is possible that when using a fixed repetition scheme those in the partial ROM at LML were within closer proximities to task failure than those in the full ROM group. Although entirely speculative, being that full ROM also includes the lengthened portion of the ROM, it would be noteworthy to see

whether such differences in hypertrophy would remain if exercise was prescribed to task failure.

Similar to the Pedrosa et al. [51] investigation, Kassiano and co-authors employed various ROM configurations to examine the effects on muscle hypertrophy outcomes of the gastrocnemius in non-resistance trained individuals [22]. Of the three ROMs, one group performed bilateral calf-raise using a leg press machine with a partial ROM at LML (defined as a starting ankle angle –25° and end ankle angle 0° [0° being foot 90° relative to tibia]), while another group performed the same exercise with a full ROM (defined as a starting ankle angle –25° and end ankle angle 25°) [22]. These groups performed 3 sets of 15–20RM, relative to each ROM, thrice weekly for 8 weeks. Following the intervention, authors reported significantly greater increases in medial gastrocnemius muscle thickness (taken via B-mode ultrasound; site was defined as the thickest and most prominent site of the leg from a posteroanterior view) for the partial ROM at LML group compared to full ROM. However, changes in lateral gastrocnemius muscle thickness (proximal third distance between the lateral epicondyle of the femur and lateral malleolus of the fibula) were not statistically different between groups. This study perhaps is the most compelling body of evidence to support the suggestion that partial ROM at LML supersedes full ROM for muscle hypertrophy as it may be the case that the gastrocnemius experiences active insufficiency (i.e., when a multi-joint muscle shortens over both joints simultaneously creating immense slack where tension experienced may be impaired) when training at SML [32], however, there remains some caution when extrapolating such data. For example, muscle size was only measured at the one site for both medial and lateral heads of the gastrocnemius, which deters our ability to observe any potential changes or difference in changes at more proximal or distal sites. A study conducted by Goto et al. [14] has previously been used as evidence to support the notion that a partial ROM at LML leads to greater muscle hypertrophy than full ROM training [71]. Interestingly, the authors of the original paper discuss the partial ROM group as primarily performing the midportion of the ROM (defined as a starting elbow angle 45° [0° being full elbow extension] and end elbow angle 90°), or in other words a partial ROM at moderate muscle lengths (MML) [14]; while the full ROM group who performed exercise from a starting elbow angle 0° and end elbow angle 120°, was the only group to include the initial or more lengthened portions of the ROM. These two groups performed the lying barbell elbow extension exercise three times per week for 8 weeks, with sessions consisting of 3 sets of an 8RM (i.e., exercise performed to task failure), relative to each ROM. Authors estimated triceps brachii muscle CSA via multiplying the muscle thickness and arm circumference

of the triceps brachii at 60% proximal distance between the acromion and olecranon. Following the 8-week intervention, authors observed significant increases in triceps brachii muscle CSA for both full ROM and partial ROM at MML, however, increases for partial ROM at MML were significantly greater than full ROM [14]. Authors attributed such differences in muscle hypertrophy to be explained by a greater degree of intramuscular hypoxia induced via partial ROM at MML compared to full ROM. However, as we have discussed in a previous section it is less likely that greater levels of hypoxia may be playing any substantial role in promoting greater hypertrophy when protocols are performed to or near fatigue as done in this study. Other factors that may explain these findings may be related to the ability to maintain higher levels of muscle activation and greater mechanical tension experienced in the middle portion of the ROM [40]. Nonetheless, despite demonstrating that performing the lying barbell elbow extension exercise with a partial ROM at MML leads to greater increases in muscle size than a full ROM, these findings should be taken with caution and considered as preliminary, as several methodological considerations should be noted when interpreting the results. For example, it is possible that estimating muscle CSA via multiplying muscle thickness and arm circumference values may have contributed to greater amounts of error than if simply utilizing the muscle thickness data or having used MRI for muscle CSA measures, as both groups experienced an extraordinarily large increase in estimated muscle CSA in habitually trained individuals (~49% increase for partial ROM at MML and ~28% increase for the full ROM). The inclusion of a time-matched, non-exercise control group may have aided in contextualizing the magnitude of muscle hypertrophy observed and random noise associated with such measuring technique.

Overall, these bodies of evidence are not entirely compelling on their own and lack the consistency in findings to support the claim that partial ROM at LML is superior to full ROM for muscle hypertrophy. However, the current authors would like to make it clear that we are not discrediting these lines of evidence but instead wish to illustrate to the reader that there is very little data directly comparing these two ROMs and much more research is needed prior to reaching a consensus. Future sections will examine other bodies of evidence that have been used to support the notion that partial ROM at LML may be a potentially advantageous approach to training for hypertrophy.

Studies Comparing Partial ROM at LML to Partial ROM at SML

While the data directly comparing partial ROM at LML and full ROM are preliminary and rather unconvincing on

their own, much of the support for the superiority of partial ROM at LML training is being drawn from other bodies of evidence that have compared training with partial ROM at LML to partial ROM at SML. McMahon et al. [35] compared performing a variety of lower body exercises with either a partial ROM at LML (defined as a starting knee angle 40° [0° being full extension] and end knee angle 90°) or partial ROM at SML (defined as a starting knee angle 0° and end knee angle 50°) in non-resistance trained individuals. The lower body training program included a variety of free-weight, machine, and body-weight exercise and were performed thrice weekly (2 supervised and 1 unsupervised) for 8 weeks with the sets, repetitions, load, and rest intervals differing depending on the exercises and day (full program can be found in Table 2 of McMahon et al. [35]). For muscle hypertrophy outcomes, authors measured muscle anatomical CSA of the vastus lateralis via B-mode ultrasound at 25%, 50% and 75% the distance between the greater trochanter to the central palpable point between the femur and tibia heads when the knee was flexed at 90°. Following the 8-week intervention, authors observed significantly greater increases at the 75% site for the partial ROM at LML group compared to partial ROM at SML, however, no statistical differences were observed between groups for both medial and proximal sites [35]. It is possible that such findings are explained by the length-tension relationship of the quadriceps. However, other possible factors that may have influenced the magnitude of hypertrophy observed at more distal sites may be related to differences in regional muscle activation between groups (in favor of the partial ROM at LML group) [43, 69].

Sato and colleagues [57] compared performing unilateral preacher curl exercise with either partial ROM at LML (defined as a starting elbow angle 0° [0° being full elbow extension] and end elbow angle 50°) or partial ROM at SML (defined as a starting elbow angle 80° and end elbow angle 130°) in non-resistance trained individuals. Exercise was performed twice weekly for 5 weeks with sessions (depending on the day) consisting of 3 sets of 10 repetitions at 30–100% maximal voluntary contraction (MVC)-isometric torque relative to ROM (linear progression in % MVC across sessions, full program can be found in methods section of Sato et al. [57]). Important to note, proximity to failure for each respective group was not specified by authors as they used a fixed repetition scheme for all participants regardless of potential individual differences in local muscular endurance, specifically for the earlier sessions corresponding to submaximal loadings. Biceps brachii muscle thickness was measured via B-mode ultrasound at 50%, 60% and 70% the distance between the lateral epicondyle of the humerus and the acromion. Authors observed significantly greater increases in muscle thickness at the 70% site for the partial

ROM at LML compared to the partial ROM at SML group. However, there were no differences observed between groups at either 50% or 60% muscle thickness sites [57]. In a similar design, Pedrosa and colleagues [52] observed almost identical findings when comparing the unilateral dumbbell preacher curl with either a partial ROM at LML (defined as a starting elbow angle 0° [0° being full elbow extension] and end elbow angle 68°) or partial ROM at SML (defined as a starting elbow angle 68° and end elbow angle 135°) in non-resistance trained individuals. These authors observed significantly greater increases in distal muscle hypertrophy of the biceps brachii (70% the distance between the acromion to the lateral epicondyle of the humerus) for the partial ROM at LML condition compared to the partial ROM at SML condition. However, there were no statistical differences between groups for proximal muscle hypertrophy (50% the distance between the acromion to the lateral epicondyle of the humerus). These findings are interesting, as they seemingly go against the length-tension theory as the biceps brachii have been suggested to be more active on the ascending limb of the length-tension curve [23], meaning they may not benefit any greater from training at LML compared to SML. However, one possible explanation for such findings may be explained by the exercise performed for these two studies (i.e., the dumbbell preacher curl exercise). There is some acute data to suggest that the biceps brachii produce greater levels of force and experience greater levels of muscle activation at the beginning portion of the dumbbell preacher curl exercise, while force production and muscle activation levels drop off towards the end of the concentric range of motion [41, 46]. Therefore, it is possible that results may be explained more so by exercise selection and the resistance profile of the biceps brachii than simply training exclusively at LML. Future studies comparing different exercises and manipulating the ROM of the biceps brachii are needed to test such a thesis.

The remaining two studies that included a comparison between partial ROM at LML training and partial ROM at SML training have been discussed previously [22, 51]. In brief, the Pedrosa et al. [51] investigation favored partial ROM at LML to partial ROM at SML for both rectus femoris and vastus lateralis muscle hypertrophy. However, the differences between groups were much more drastic than the differences between partial ROM at LML compared to full ROM, as the partial ROM at SML only observed significant hypertrophy of the more proximal regions of both the muscles measured (40% and 50% sites) and no significant changes at more distal sites (60% and 70% sites), while partial ROM at LML increased significantly at all sites for both muscles [51]. Apart from the inherent limitation of accounting for differences in local muscular endurance as discussed above, it is very likely that the ROM used for partial ROM

at SML (defined as a starting knee angle 65° and end knee angle 30°) simply did not activate more distal regions of the muscle and overtime resulted in no significant changes in muscle CSA.

In the Kassiano et al. investigation [22], authors also allowed for a comparison between partial ROM at LML and partial ROM at SML (defined as a starting ankle angle 0° and end ankle angle 25°) when investigating changes in gastrocnemius muscle thickness. Changes in medial gastrocnemius muscle thickness were significantly greater for the partial ROM at LML group compared to the partial ROM at SML group, with the latter of the two observing no significant hypertrophy after 8 weeks. While both groups observed significant hypertrophy for the lateral gastrocnemius, changes were significantly greater for the partial ROM at LML group compared to the partial ROM at SML group [22]. Again, these findings may be indicative of the fact that the gastrocnemius muscles experience active insufficiency at SML and thus prohibits the ability to generate a sufficient stimulus for muscle hypertrophy [32].

Overall, when synthesizing these bodies of evidence, it appears that training with a partial ROM at LML leads to significantly greater regional hypertrophy compared to partial ROM at SML (specifically more distal sites). It would be interesting to determine whether such changes reflect greater global changes across the whole muscle belly, being that all the aforementioned studies measured muscle hypertrophy with B-mode ultrasound (1-dimensional image). For example, if we take from the concentric-only versus eccentric-only literature, there is some evidence to suggest that despite differences in regional hypertrophy when measuring with B-mode ultrasound (i.e., greater proximal hypertrophy for concentric-only while eccentric-only elicits greater distal hypertrophy) [12], MRI changes in whole muscle volume changes are similar between protocols [4, 12]. It would be interesting to see if such phenomena would be corroborated when training with these two distinct ROMs.

Drawing from Isometrics

While the purpose of this manuscript was to examine studies that have manipulated ROM during resistance training interventions, of relevance is a review paper by Oranchuk et al. [47] examining isometric studies that is commonly cited as evidence for training exclusively at LML. Authors observed that isometric training at LML, often led to significantly greater increases in muscle size than isometric training at SML. However, what most authors fail to reconcile is that all studies that were discussed by Oranchuk et al. [47] were conducted in the quadriceps [2, 25, 45], again adding to the question as to whether such findings are then generalizable to all muscle groups as performing isometric

exercise at LML in the tibialis anterior [1] and hamstrings [42] have not resulted in greater hypertrophic adaptations when compared to SML. Additionally, the studies were all volume-matched, which raises a similar concern as to whether those in the LML training groups were more apt to fatigue and necessitate a higher degree of motor unit recruitment across the target musculature, which seems important for protocols that elicit significant hypertrophy [8, 39], than those in the SML groups. Nonetheless, these studies may be used as rationale to investigate whether similar differences may transpire when comparing partial ROM at LML to partial ROM at SML but may not be used as direct evidence for the superiority of partial ROM at LML being that isometric studies do not manipulate ROM per se.

Conclusions

It is believed that, depending on the characteristics of muscles (e.g., force producing capabilities throughout distinct joint angles), manipulating ROM may play a significant role in the manifestations of muscle hypertrophic adaptations to training [43, 48]. With this in mind, authors of recent review papers have suggested that partial ROM at LML may be a potential method to prescribe during training to maximize muscle hypertrophy instead of other ROM configurations [3, 71]. Being that statements from meta-analyses and review papers often guide practical recommendations for exercise programming, the goal of the present manuscript was to evaluate the evidence for the potential superiority of partial ROM at LML and possible physiological mechanisms underpinning such phenomena. While it appears that partial ROM at LML is an effective means to promote skeletal muscle hypertrophy and inherently no downside to this style of training (i.e., at least comparable muscle hypertrophy to full ROM), there is little data to suggest that this method of training is superior to full ROM training for muscle hypertrophy, and it is less known whether the physiologic rationale for performing exclusively partial ROM at LML may apply to all muscle groups, being that majority of the studies to date have been isolated to the quadriceps muscles. While we are open to the possibility of partial ROM at LML indeed being a potentially more advantageous method for maximizing muscle hypertrophy compared to other ROMs, we are not compelled by the existing data to warrant such a statement. Furthermore, there are a bevy of considerations to be made as this area of interest expands in the coming years which may involve investigating the long-term effects of performing partial ROM exercise at either LML or SML on sports performance and injury rehabilitation outcomes as it is not well-known whether there are any potential downsides (or upsides) from eliminating portions of ROM for a

given exercise. Lastly, a large majority of the studies to date have involved non-resistance trained cohorts and therefore such data is not necessarily able to be extrapolated to resistance-trained individuals requiring future inquiries in trained cohorts. Future studies should focus on comparing various ROM configurations on a variety of muscle groups, populations (e.g., athletic populations), and include a variety of exercises to determine whether “optimal ROM” is influenced by such factors.

Author Contributions ENM and SLB conceived the study idea for this article; ENM wrote the first draft; SLB, WAAC, SLG, and HTB suggested critical revisions. All authors have read and approved the final version of the manuscript and agree with the order of presentation of the authors.

Declarations

Conflict of interest SLB, ENM, WAAC and SLG declare that they have no conflicts of interest that are relevant to the content of this article. HTB is owner of a nutrition and training education company.

References

1. Akagi R, Hinks A, Power GA. Differential changes in muscle architecture and neuromuscular fatigability induced by isometric resistance training at short and long muscle-tendon unit lengths. *J Appl Physiol.* 2020;129(1):173–84.
2. Alegre LM, Ferri-Morales A, Rodriguez-Casares R, Aguado X. Effects of isometric training on the knee extensor moment–angle relationship and vastus lateralis muscle architecture. *Eur J Appl Physiol.* 2014;114(11):2437–46.
3. Androulakis Korakakis P, Wolf M, Coleman M, Burke R, Piñero A, Nippard J, Schoenfeld BJ. Optimizing resistance training technique to maximize muscle hypertrophy: a narrative review. *J Funct Morphol Kinesiol.* 2023;9(1):9.
4. Blazevich AJ, Cannavan D, Coleman DR, Horne S. Influence of concentric and eccentric resistance training on architectural adaptation in human quadriceps muscles. *J Appl Physiol.* 2007;103(5):1565–75.
5. Bloomquist K, Langberg H, Karlsen S, Madsgaard S, Boesen M, Raastad T. Effect of range of motion in heavy load squatting on muscle and tendon adaptations. *Eur J Appl Physiol.* 2013;113(8):2133–42.
6. Burkholder TJ. Mechanotransduction in skeletal muscle. *Front Biosci.* 2007;12:174.
7. Dankel SJ, Mattocks KT, Jessee MB, Buckner SL, Mouser JG, Loenneke JP. Do metabolites that are produced during resistance exercise enhance muscle hypertrophy? *Eur J Appl Physiol.* 2017;117(11):2125–35.
8. Dankel SJ, Jessee MB, Mattocks KT, Mouser JG, Counts BR, Buckner SL, Loenneke JP. Training to fatigue: the answer for standardization when assessing muscle hypertrophy? *Sports Med.* 2017;47(6):1021–7.
9. Farup J, De Paoli F, Bjerg K, Riis S, Ringgaard S, Vissing K. Blood flow restricted and traditional resistance training performed to fatigue produce equal muscle hypertrophy. *Scand J Med Sci Sports.* 2015;25(6):754–63.

10. Fink JE, Schoenfeld BJ, Kikuchi N, Nakazato K. Acute and long-term responses to different rest intervals in low-load resistance training. *Int J Sports Med*. 2017;38(2):118–24.
11. Fitch S, McCOMAS A. Influence of human muscle length on fatigue. *J Physiol*. 1985;362:205–13.
12. Franchi MV, Atherton PJ, Reeves ND, Flück M, Williams J, Mitchell WK, Selby A, Beltran Valls R, Narici MV. Architectural, functional and molecular responses to concentric and eccentric loading in human skeletal muscle. *Acta Physiol*. 2014;210(3):642–54.
13. Gordon A, Huxley AF, Julian F. The variation in isometric tension with sarcomere length in vertebrate muscle fibres. *J Physiol*. 1966;184(1):170–92.
14. Goto M, Maeda C, Hirayama T, Terada S, Nirengi S, Kurosawa Y, Nagano A, Hamaoka T. Partial range of motion exercise is effective for facilitating muscle hypertrophy and function through sustained intramuscular hypoxia in young trained men. *J Strength Cond Res*. 2019;33(5):1286–94.
15. Heaselgrave SR, Blacker J, Smeuninx B, McKendry J, Breen L. Dose-response relationship of weekly resistance-training volume and frequency on muscular adaptations in trained men. *Int J Sports Physiol Perform*. 2019;14(3):360–8.
16. Herzog W. Mechanisms of enhanced force production in lengthening (eccentric) muscle contractions. *J Appl Physiol*. 2014;116(11):1407–17.
17. Herzog W. Why are muscles strong, and why do they require little energy in eccentric action? *J Sport Health Sci*. 2018;7(3):255–64.
18. Hodson N, West DW, Philp A, Burd NA, Moore DR. Molecular regulation of human skeletal muscle protein synthesis in response to exercise and nutrients: a compass for overcoming age-related anabolic resistance. *Am J Physiol Cell Physiol*. 2019;317(6):C1061–78.
19. Joannis S, Lim C, McKendry J, Mcleod JC, Stokes T, Phillips SM. Recent advances in understanding resistance exercise training-induced skeletal muscle hypertrophy in humans. *F1000Res*. 2020;9:F1000.
20. Jorgenson KW, Phillips SM, Hornberger TA. Identifying the structural adaptations that drive the mechanical load-induced growth of skeletal muscle: a scoping review. *Cells*. 2020;9(7):1658.
21. Kassiano W, Costa B, Nunes JP, Ribeiro AS, Schoenfeld BJ, Cyrino ES. Which ROMs lead to Rome? A systematic review of the effects of range of motion on muscle hypertrophy. *J Strength Cond Res*. 2022;37(5):1135–44.
22. Kassiano W, Costa B, Kunevaliki G, Soares D, Zacarias G, Manske I, Takaki Y, Ruggiero MF, Stavinski N, Francsuel J. Greater gastrocnemius muscle hypertrophy after partial range of motion training performed at long muscle lengths. *J Strength Cond Res*. 2023;37(9):1746–53.
23. Koo TK, Mak AF, Hung L. In vivo determination of subject-specific musculotendon parameters: applications to the prime elbow flexors in normal and hemiparetic subjects. *Clin Biomech*. 2002;17(5):390–9.
24. Kooistra R, Blaauw M, Born J, De Ruijter C, De Haan A. Knee extensor muscle oxygen consumption in relation to muscle activation. *Eur J Appl Physiol*. 2006;98(6):535–45.
25. Kubo K, Ohgo K, Takeishi R, Yoshinaga K, Tsunoda N, Kanehisa H, Fukunaga T. Effects of isometric training at different knee angles on the muscle–tendon complex in vivo. *Scand J Med Sci Sports*. 2006;16(3):159–67.
26. Kubo K, Ikebukuro T, Yata H. Effects of squat training with different depths on lower limb muscle volumes. *Eur J Appl Physiol*. 2019;119(9):1933–42.
27. Laurentino G, Ugrinowitsch C, Aihara A, Fernandes A, Parcell A, Ricard M, Tricoli V. Effects of strength training and vascular occlusion. *Int J Sports Med*. 2008;29(8):664–7.
28. Laurentino GC, Ugrinowitsch C, Roschel H, Aoki MS, Soares AG, MANOEL NEVES J, Aihara AY, Fernandes AdRC, and Tricoli V. Strength training with blood flow restriction diminishes myostatin gene expression. *Med Sci Sports Exerc*. 2012;44(3):406–12.
29. Lieber RL, Roberts TJ, Blemker SS, Lee SS, Herzog W. Skeletal muscle mechanics, energetics and plasticity. *J Neuroeng Rehabil*. 2017;14(1):108.
30. Lim C, Nunes EA, Currier BS, Mcleod JC, Thomas AC, Phillips SM. An evidence-based narrative review of mechanisms of resistance exercise-induced human skeletal muscle hypertrophy. *Med Sci Sports Exerc*. 2022;54(9):1546–59.
31. MacDougall J, Ray S, Sale D, McCartney N, Lee P, Garner S. Muscle substrate utilization and lactate production. *Can J Appl Physiol*. 1999;24(3):209–15.
32. Maganaris CN. Force-length characteristics of the in vivo human gastrocnemius muscle. *Clin Anat*. 2003;16(3):215–23.
33. Marcotte GR, West DW, Baar K. The molecular basis for load-induced skeletal muscle hypertrophy. *Calcif Tissue Int*. 2015;96(3):196–210.
34. McKendry J, Stokes T, Mcleod JC, Phillips SM. Resistance exercise, aging, disuse, and muscle protein metabolism. *Compr Physiol*. 2021;11(3):2249–78.
35. McMahon G, Morse CI, Burden A, Winwood K, Onambélé GL. Muscular adaptations and insulin-like growth factor-1 responses to resistance training are stretch-mediated. *Muscle Nerve*. 2014;49(1):108–19.
36. McMahon GE, Morse CI, Burden A, Winwood K, Onambélé GL. Impact of range of motion during ecologically valid resistance training protocols on muscle size, subcutaneous fat, and strength. *J Strength Cond Res*. 2014;28(1):245–55.
37. Mitchell CJ, Churchward-Venne TA, West DW, Burd NA, Breen L, Baker SK, Phillips SM. Resistance exercise load does not determine training-mediated hypertrophic gains in young men. *J Appl Physiol*. 2012;113(1):71–7.
38. Morton RW, Oikawa SY, Wavell CG, Mazara N, McGlory C, Quadriatero J, Baechler BL, Baker SK, Phillips SM. Neither load nor systemic hormones determine resistance training-mediated hypertrophy or strength gains in resistance-trained young men. *J Appl Physiol*. 2016;121(1):129–38.
39. Morton RW, Colenso-Semple L, Phillips SM. Training for strength and hypertrophy: an evidence-based approach. *Curr Opin Physiol*. 2019;10(1):90–5.
40. Murray WM, Buchanan TS, Delp SL. The isometric functional capacity of muscles that cross the elbow. *J Biomech*. 2000;33(8):943–52.
41. Murray WM, Buchanan TS, Delp SL. Scaling of peak moment arms of elbow muscles with upper extremity bone dimensions. *J Biomech*. 2002;35(1):19–26.
42. Nakao S, Ikezoe T, Taniguchi M, Motomura Y, Hirono T, Nojiri S, Hayashi R, Tanaka H, Ichihashi N. Effects of low-intensity torque-matched isometric training at long and short muscle lengths of the hamstrings on muscle strength and hypertrophy: a randomized controlled study. *J Strength Cond Res*. 2023;37(10):1978–84.
43. Newmire DE, Willoughby DS. Partial compared with full range of motion resistance training for muscle hypertrophy: a brief review and an identification of potential mechanisms. *J Strength Cond Res*. 2018;32(9):2652–64.
44. Ng A, Agre J, Hanson P, Harrington M, Nagle F. Influence of muscle length and force on endurance and pressor responses to isometric exercise. *J Appl Physiol*. 1994;76(6):2561–9.
45. Noorkoiv M, Nosaka K, Blazevich AJ. Neuromuscular adaptations associated with knee joint angle-specific force change. *Med Sci Sports Exerc*. 2014;46(8):1525–37.
46. Oliveira LF, Matta TT, Alves DS, Garcia MA, Vieira TM. Effect of the shoulder position on the biceps brachii EMG in different dumbbell curls. *J Sci Med Sport*. 2009;8(1):24.

47. Oranchuk DJ, Storey AG, Nelson AR, Cronin JB. Isometric training and long-term adaptations: effects of muscle length, intensity, and intent: a systematic review. *Scand J Med Sci Sports*. 2019;29(5):484–503.
48. Ottinger CR, Sharp MH, Stefan MW, Gheith RH, de la Escriella F, Wilson JM. Muscle hypertrophy response to range of motion in strength training: a novel approach to understanding the findings. *Strength Cond J*. 2023;45(2):162–76.
49. Ozaki H, Loenneke JP, Buckner SL, Abe T. Muscle growth across a variety of exercise modalities and intensities: contributions of mechanical and metabolic stimuli. *Med Hypotheses*. 2016;88:22–6.
50. Pallarés JG, Hernández-Belmonte A, Martínez-Cava A, Vetrovsky T, Steffl M, Courel-Ibáñez J. Effects of range of motion on resistance training adaptations: a systematic review and meta-analysis. *Scand J Med Sci Sports*. 2021;31(10):1866–81.
51. Pedrosa GF, Lima FV, Schoenfeld BJ, Lacerda LT, Simões MG, Pereira MR, Diniz RC, Chagas MH. Partial range of motion training elicits favorable improvements in muscular adaptations when carried out at long muscle lengths. *Eur J Sport Sci*. 2022;22(8):1250–60.
52. Pedrosa GF, Simões MG, Figueiredo MO, Lacerda LT, Schoenfeld BJ, Lima FV, Chagas MH, Diniz RC. Training in the initial range of motion promotes greater muscle adaptations than at final in the arm curl. *Sports*. 2023;11(2):39.
53. Phillips SM. A brief review of critical processes in exercise-induced muscular hypertrophy. *Sports Med*. 2014;44(Suppl 1):71–7.
54. Pinto RS, Gomes N, Radaelli R, Botton CE, Brown LE, Bottaro M. Effect of range of motion on muscle strength and thickness. *J Strength Cond Res*. 2012;26(8):2140–5.
55. Rasmussen BB, Phillips SM. Contractile and nutritional regulation of human muscle growth. *Exerc Sport Sci Rev*. 2003;31(3):127–31.
56. Rindom E, Vissing K. Mechanosensitive molecular networks involved in transducing resistance exercise-signals into muscle protein accretion. *Front Physiol*. 2016;7:547.
57. Sato S, Yoshida R, Kiyono R, Yahata K, Yasaka K, Nunes JP, Nosaka K, Nakamura M. Elbow joint angles in elbow flexor unilateral resistance exercise training determine its effects on muscle strength and thickness of trained and non-trained arms. *Front Physiol*. 2021;12:734509.
58. Schoenfeld BJ. Potential mechanisms for a role of metabolic stress in hypertrophic adaptations to resistance training. *Sports Med*. 2013;43(3):179–94.
59. Schoenfeld BJ, Grgic J. Effects of range of motion on muscle development during resistance training interventions: a systematic review. *SAGE Open Med*. 2020;8:2050312120901559.
60. Schoenfeld BJ, Pope ZK, Benik FM, Hester GM, Sellers J, Nooner JL, Schnaiter JA, Bond-Williams KE, Carter AS, Ross CL. Longer inter-set rest periods enhance muscle strength and hypertrophy in resistance-trained men. *J Strength Cond Res*. 2016;30(7):1805–12.
61. Schoenfeld BJ, Contreras B, Krieger J, Grgic J, Delcastillo K, Belliard R, Alto A. Resistance training volume enhances muscle hypertrophy but not strength in trained men. *Med Sci Sports Exerc*. 2019;51(1):94.
62. Schott J, McCully K, Rutherford O. The role of metabolites in strength training: II. Short versus long isometric contractions. *Eur J Appl Physiol Occup Physiol*. 1995;71(4):337–41.
63. Son J, Indresano A, Sheppard K, Ward SR, Lieber RL. Intraoperative and biomechanical studies of human vastus lateralis and vastus medialis sarcomere length operating range. *J Biomech*. 2018;67:91–7.
64. Street SF. Lateral transmission of tension in frog myofibers: a myofibrillar network and transverse cytoskeletal connections are possible transmitters. *J Cell Physiol*. 1983;114(3):346–64.
65. Teixeira EL, Ugrinowitsch C, de Salles Painelli V, Silva-Batista C, Aihara AY, Cardoso FN, Roschel H, Tricoli V. Blood flow restriction does not promote additional effects on muscle adaptations when combined with high-load resistance training regardless of blood flow restriction protocol. *J Strength Cond Res*. 2021;35(5):1194–200.
66. Tesch PA, Colliander EB, Kaiser P. Muscle metabolism during intense, heavy-resistance exercise. *Eur J Appl Physiol Occup Physiol*. 1986;55(4):362–6.
67. Valamatos MJ, Tavares F, Santos RM, Veloso AP, Mil-Homens P. Influence of full range of motion vs. equalized partial range of motion training on muscle architecture and mechanical properties. *Eur J Appl Physiol*. 2018;118(9):1969–83.
68. Wackerhage H, Schoenfeld BJ, Hamilton DL, Lehti M, Hulmi JJ. Stimuli and sensors that initiate skeletal muscle hypertrophy following resistance exercise. *J Appl Physiol*. 2019;126(1):30–43.
69. Wakahara T, Fukutani A, Kawakami Y, Yanai T. Nonuniform muscle hypertrophy: its relation to muscle activation in training session. *Med Sci Sports Exerc*. 2013;45(11):2158–65.
70. Werkhausen A, Solberg E, Paulsen C, Bojsen-Møller G J, and, Seynnes OR. Adaptations to explosive resistance training with partial range of motion are not inferior to full range of motion. *Scand J Med Sci Sports*. 2021;31(5):1026–35.
71. Wolf M, Androulakis-Korakakis P, Fisher J, Schoenfeld B, Steele J. Partial vs full range of motion resistance training: a systematic review and meta-analysis. *Int J Strength Cond*. 2023. <https://doi.org/10.51224/SRXIV.198>.
72. You J-S, McNally RM, Jacobs BL, Privett RE, Gundermann DM, Lin K-H, Steinert ND, Goodman CA, Hornberger TA. The role of raptor in the mechanical load-induced regulation of mTOR signaling, protein synthesis, and skeletal muscle hypertrophy. *FASEB J*. 2019;33(3):4021.
73. Zabaleta-Korta A, Fernández-Peña E, Santos-Concejero J. Regional hypertrophy, the inhomogeneous muscle growth: a systematic review. *Strength Cond J*. 2020;42(1):94–101.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.